



The oldest fossil evidence of animal parasitism by fungi supports a Cretaceous diversification of fungal–arthropod symbioses

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ABSTRACT

Paleoophiocordyceps coccophagus, a fungal parasite of a scale insect from the Early Cretaceous (Upper Albian), is reported and described here. This fossil not only provides the oldest fossil evidence of animal parasitism by fungi but also contains morphological features similar to asexual states of *Hirsutella* and *Hymenostilbe* of the extant genus *Ophiocordyceps* (Ophiocordycipitaceae, Hypocreales, Sordariomycetes, Pezizomycotina, Ascomycota). Because species of Hypocreales collectively exhibit a broad range of nutritional modes and symbioses involving plants, animals and other fungi, we conducted ancestral host reconstruction coupled with phylogenetic dating analyses calibrated with *P. coccophagus*. These results support a plant-based ancestral nutritional mode for Hypocreales, which then diversified ecologically through a dynamic process of intra- and interkingdom host shifts involving fungal, higher plant and animal hosts. This is especially evident in the families Cordycipitaceae, Clavicipitaceae and Ophiocordycipitaceae, which are characterized by a high occurrence of insect pathogens. The ancestral ecologies of Clavicipitaceae and Ophiocordycipitaceae are inferred to be animal pathogens, a trait inherited from a common ancestor, whereas the ancestral host affiliation of Cordycipitaceae was not resolved. Phylogenetic dating supports both a Jurassic origin of fungal–animal symbioses within Hypocreales and parallel diversification of all three insect pathogenic families during the Cretaceous, concurrent with the diversification of insects and angiosperms.

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1. Introduction

Species of Hypocreales form diverse symbiotic associations that include antagonistic to mutualistic interactions with numerous animals, plants and other fungi. As such, they are a unique clade among fungi and provide a model system for understanding evolutionary processes of host affiliation (Rossman et al., 1999; Spatafora et al., 2007). Animal associations are primarily represented by parasites and pathogens, or more rarely mutualistic endosymbionts, of arthropods (Spatafora et al., 2007; Suh et al., 2001; Sung et al., 2007a). Plant associations include decomposers (e.g., *Bionectria*) and pathogens (e.g., *Fusarium*), with a preponderance of associations with grasses where they can function as epiphytes (e.g., *Claviceps* on rye) or endophytes (e.g., *Neotyphodium* on tall fescue) in both antagonistic and beneficial associations (Clay and Schardl, 2002; Rossman et al., 1999). In associations with other fungi, they function as mycoparasites that infect various groups of fungi including mushrooms, bracket fungi, rusts and truffles (Currie et al., 2003; Rossman et al., 1999; Spatafora et al., 2007).

The current pattern of host affiliation of hypocrealean fungi is inferred to be an evolutionary product of intra- and interkingdom host shifts with one of the most dramatic examples being the origin of grass symbionts (e.g., *Claviceps* and *Epichloë*) from animal pathogens (Spatafora et al., 2007). The most common hosts of the hypocrealean animal pathogens include species of Coleoptera, Hemiptera and Lepidoptera, although any one species of arthropod pathogen is generally considered to have a narrow host range of one or closely related host species (Kobayasi, 1941, 1982; Mains, 1958). These arthropod pathogenic fungi consist of several genera (e.g., *Cordyceps* s. s., *Elaphocordyceps*, *Hypocrella*, *Metacordyceps* and *Ophiocordyceps*) in three families (Clavicipitaceae, Cordycipitaceae and Ophiocordycipitaceae) (Sung et al., 2007a). These multiple fungal lineages are associated with a diversity of arthropods that possibly had a relatively ancient origin. Therefore, knowing when these diverse symbioses appeared in geologic time is central to understanding the evolution of Hypocreales.

Here, we report an Early Cretaceous Burmese amber male scale insect parasitized by a fungus with a striking morphological similarity to *Hirsutella* and *Hymenostilbe* asexual states of *Ophiocordyceps* (Ophiocordycipitaceae, Hypocreales, Sordariomycetes, Pezizomycotina, Ascomycota). In fungi, numerous species are pleomorphic and produce more than one spore-producing state in their

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life cycle. The different spore-producing states may or may not co-exist spatially or temporally and the linkage of meiotic and mitotic spore-producing states to a common life cycle is often unknown. Article 59 of the International Botanical Code (McNeill et al., 2006) allows for unique Latin binomials for meiotic (e.g., *Ophiocordyceps*) and mitotic (e.g., *Hirsutella* and *Hymenostilbe*) stages of the life cycles of Ascomycota and Basidiomycota. Recent phylogenetic analyses have demonstrated that many asexual states are phylogenetically informative for the Hypocreales and that *Hirsutella sensu stricto* and *Hymenostilbe* are restricted to the genus *Ophiocordyceps* (Sung et al., 2007a,b).

The discovery of a new fossil record in this study is not only the earliest fossil record of animal parasitism by fungi, but it also provides an opportunity to estimate the divergence times of the major lineages of Hypocreales. Therefore, we use this fossil to calibrate molecular clock analyses of the Hypocreales based on multiple genes and provide the first example of Cretaceous diversification of numerous, closely related lineages of Fungi. The estimated dates are compared to the fossil record of their symbiotic partners in order to correlate estimated divergence times with their major intra- and interkingdom host shifts (Farrell, 1998; Percy et al., 2004). As such, the main objectives of this study are to (1) describe the newly discovered Burmese amber fossil, (2) estimate the divergence times of the major lineages of Hypocreales and (3) further develop hypotheses for ancestral character states, diversification and the geologic origin of host affiliation in Hypocreales.

2. Materials and methods

2.1. Microscopy and phylogenetic sampling

To describe a new fossil record, observations and photographs were made with a Nikon SMZ-10 R stereoscopic microscope and Nikon Optiphot TM compound microscope. To construct a phylogeny of major lineages in Hypocreales and estimate their divergence times, representative taxa of members from all the major families were chosen based on previous phylogenetic studies (Castlebury et al., 2004; Sung et al., 2007a). A total of 148 taxa were selected to represent the morphological and ecological diversity of Hypocreales, including outgroup taxa (*Glomerella cingulata* and *Verticillium dahliae*). We used DNA sequences from five genes: nuclear ribosomal small and large subunit DNA (*nrSSU* and *nrLSU*), elongation factor 1 α (*TEF*), the largest and second largest subunits of RNA polymerase II (*RPB1* and *RPB2*). Most of the DNA sequences used in this study were obtained from the previous phylogenetic studies of Hypocreales (Castlebury et al., 2004; Sung et al., 2007a). A total of 11 *RPB2* sequences were newly generated for selected taxa from the study of Castlebury et al. (2004) as previously described in the study of Sung et al. (2007a). Information on taxa and sequences used in this study is presented in Supplementary Table 1.

2.2. Sequence alignment and phylogenetic analyses

DNA sequences were aligned with Clustal W (Thompson et al., 1999) using default settings, followed by optimization in BioEdit (Hall, 1999) by direct examination. The final alignment was submitted to TreeBASE (Study Accession No. = S2049, Matrix Accession No. = M3835). The ambiguously aligned regions were excluded from all phylogenetic analyses. Maximum likelihood (ML) analyses were performed with RAxML-VI-HPG v2.0 (Stamatakis et al., 2005). Different rates of substitutions were incorporated for 11 partitions of *nrSSU*, *nrLSU* and each codon of the three protein-coding genes using a GTRCAT model of evolution with 25 rate categories. Nodal support was estimated using 200 replicates of non-parametric bootstrapping (Felsenstein, 1985). Bayesian analy-

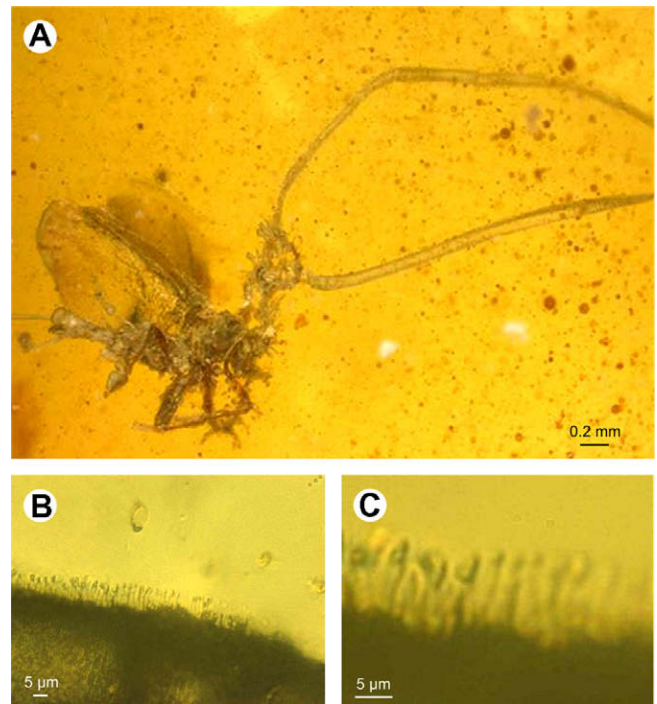


Fig. 1. Photographs of the holotype of *P. coccophagus* gen. et sp. nov. that shows the oldest evidence of fungal parasitism of animal. (A) Synnemata arising from a male scale insect (Albicoccidae) in Burmese amber. (B) Conidiogenous cells that are distributed in a hymenium-like layer. (C) Conidia and conidiogenous cells.

ses were conducted with MrBayes version 3.1 (Huelsenbeck and Ronquist, 2001) using the general time-reversible model with invariant sites and gamma distribution (GTR + I + Γ) for 11 partitions (*nrSSU*, *nrLSU* and each codon of protein-coding genes). To examine the convergence of log-likelihood, we performed five independent Bayesian analyses with random starting trees and 10,000,000 generations sampled every 1000th tree. The variation in log-likelihoods of the sampled trees was investigated by analyzing the MrBayes log file. To confirm the convergence of all five 10,000,000 generation analyses, the stability of posterior probabilities was assessed using the program AWTY (Nylander et al., 2008). After the initial 3000 trees (3,000,000 generations) were removed, the pairwise comparisons of split frequencies indicated that five 10,000,000 analyses had converged (Supplementary Fig. 2). Therefore, the initial 3000 trees were identified as burnin and the remaining 7000 trees from one of the five analyses were used to construct a 50% majority-rule consensus tree and calculate associated posterior probabilities. In interpreting phylogenetic confidence, we consider nodes strongly supported when they receive both bootstrap proportion (BP) $\geq 70\%$ and posterior probability (PP) ≥ 0.95 .

2.3. Ancestral state reconstruction and divergence time estimation

To understand evolutionary patterns of host affiliations among the hypocrealean fungi, we reconstructed the ancestral host affiliations by coding each taxon mainly based on host kingdom (e.g., animal, fungi, plant, soil) (Supplementary Table 1). We used the maximum likelihood model Mk1, as implemented in Mesquite 1.12 (Maddison and Maddison, 2006). In estimating the divergence times of early evolutionary events in the Hypocreales, we used a Bayesian relaxed molecular clock approach that does not have an assumption of a strict clock and that allows for the simultaneous use of different models for five gene loci used in this study (Kishino et al., 2001). Due to the uncertainty of the placement of the

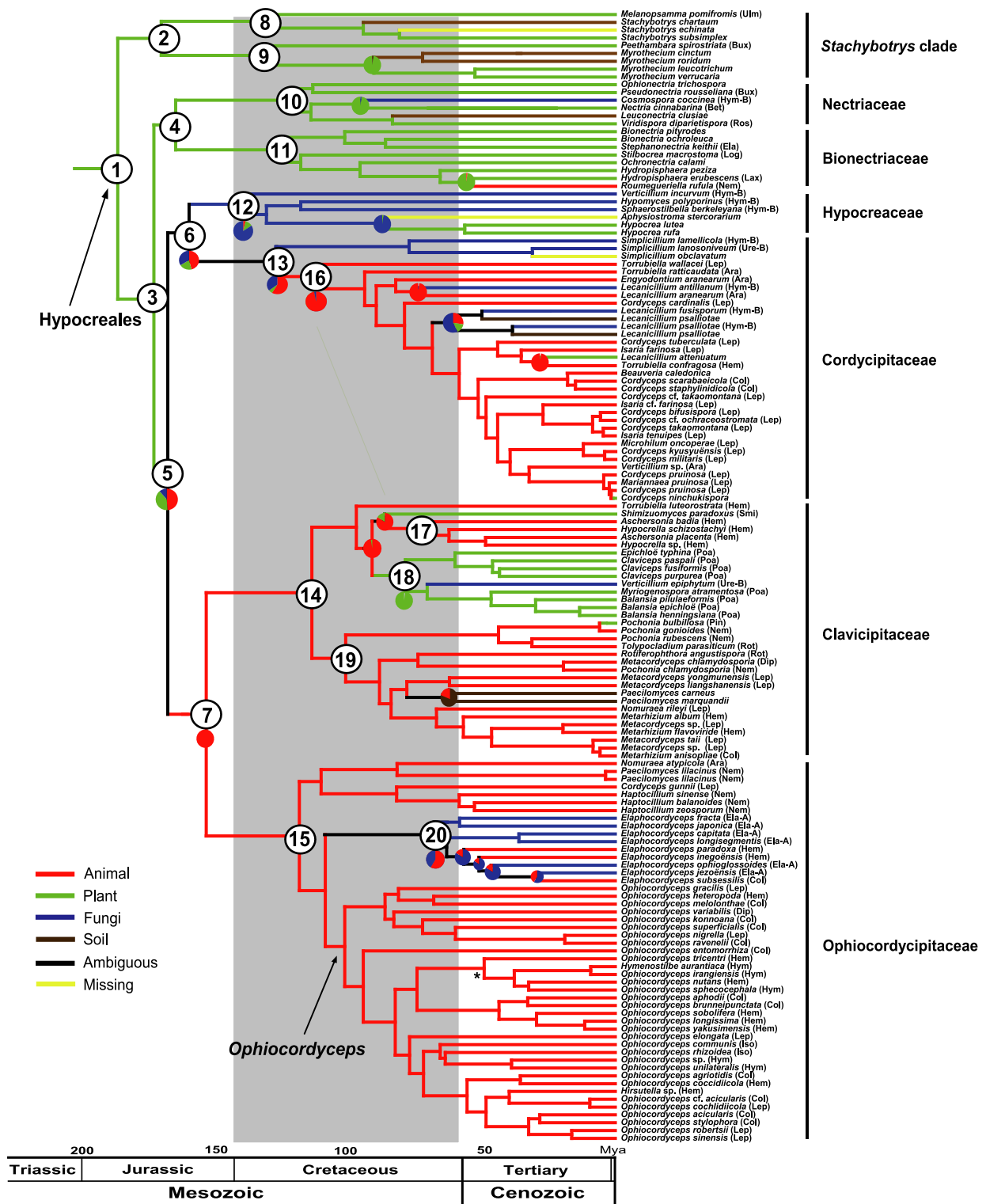


Fig. 2. Divergence age estimates of major lineages of Hypocreales. Chronogram was constructed based on the tree (Supplementary Fig. 1) from maximum likelihood analyses. Calibration (crown node of genus *Ophiocordyceps*) was based on *P. coccophagus*, a fungal parasite of a scale insect from the Early Cretaceous (Upper Albian). *Hymenostilbe* clade is marked with an asterisk (*) below the corresponding node. Geological times are provided below the chronogram and the gray rectangular box highlights the Cretaceous. Numbers in the circles correspond to twenty selected major nodes of which the divergence times are shown in Table 1. Ancestral nutritional modes are colored for the corresponding branches (black, ambiguous; blue, fungal; brown, soil; green, plant; red, animal and yellow, missing) and the proportions of likelihoods are provided using a pie chart for the corresponding nodes which are considered being important in evolution of host affiliation in hypocrealean fungi. More definitive host information (also see Supplementary Table 1) is provided with the three-letter code in the parentheses after the species name as follows: Ara, Arachnida; Bet, Betulaceae; Bux, Buxaceae; Col, Coleoptera; Dip, Diptera; Ela-A, Elaphomyces-Ascomycota; Hem, Hemiptera; Hym, Hymenoptera; Hym-B, Hymenomyces-Basidiomycota; Iso, Isoptera; Lax, Laxmanniaceae; Lep, Lepidoptera; Log, Loganiaceae; Mol, Diplopoda; Nem, Nematoda; Pin, Pinaceae; Poa, Poaceae; Ros, Rosaceae; Rot, Rotifera; Smi, Smilacaceae; Ulm, Ulmaceae; Ure-B, Uredinales-Basidiomycota.

Table 1

Bayesian estimates of divergence times (Mya), including 95% credibility intervals (CI), for the major twenty nodes in Fig. 2

Node	Age (CI)	Node	Age (CI)
1	193 (158, 232)	11	129 (101, 159)
2	176 (143, 215)	12	145 (117, 177)
3	178 (150, 213)	13	131 (104, 163)
4	170 (141, 206)	14	117 (95, 144)
5	173 (146, 206)	15	122 (109, 138)
6	165 (138, 198)	16	116 (91, 146)
7	158 (137, 187)	17	75 (54, 98)
8	135 (99, 172)	18	81 (60, 105)
9	136 (101, 176)	19	104 (83, 129)
10	125 (98, 155)	20	70 (56, 84)

Hymenostilbe clade within *Ophiocordyceps* (Sung et al., 2007a,b) and the existence of morphological intermediates between *Hirsutella* and *Hymenostilbe* (discussed below), the fossil evidence of *P. coccophagus* was used as a crown calibration point for *Ophiocordyceps* in Bayesian relaxed clock analyses (Fig. 2; Table 1). To incorporate the inherent uncertainty of the age estimate from the fossil record, the calibrated crown node was defined with lower and upper bounds of the stratigraphic range of the geological time, the Upper Albian of the Lower Cretaceous (99–105 Mya), of the Burmese amber fossil record reported in this study. Estimation of nucleotide substitutions for each gene was independently calculated with F84 + G using baseml program of PAML ver. 3.14 (Yang, 1997); these nucleotide substitution rates were then used to estimate branch lengths and the variance–covariance structure of the branch length estimates with estbranches program (<http://statgen.ncsu.edu/thorne/multidivtime.html>). The posterior ages of nodes in ingroup taxa were estimated using multidivtime program (<http://statgen.ncsu.edu/thorne/multidivtime.html>) with the prior gamma distributions on three parameters of the relaxed clock model specified (mean of root age: 200 with SD of 200, root rate: 0.0011 with SD of 0.0011, rate autocorrelation: 0.005 with SD of 0.005). We conducted five independent 1,000,000 generation analyses with sampling of every 100th generation after conservatively removing 100,000 generations as burnin. Posterior ages were nearly identical among the five analyses, consistent with the convergence of the Bayesian dating analyses. Age estimates were reported from one of the five analyses with 95% credibility intervals to address the uncertainty of the divergence time estimates.

3. Results and discussion

3.1. Paleosystematics

The oldest fossil evidence of insect parasitism by fungi was discovered in Burmese amber and described here.

Paleoophiocordyceps coccophagus G.-H. Sung, Poinar & Spatafora gen. et sp. nov.

Two synnemata of *Paleoophiocordyceps* emerge from the head of a male scale insect belonging to the family Albicoccidae Koteja (Hemiptera: Coccinea: Orthezioidea). The left synnema is approximately 2.6 mm in length and the right is approximately 2.3 mm in length (Fig. 1). Most of the surface of the synnemata is covered with a palisade of elongated cylindrical phialides 4–7 μ m in length (Fig. 1). The conidia, which range from 3 to 4 μ m in diameter, appear to be borne singly at the tip of the phialides, but polysporic phialides may exist (Fig. 1).

3.1.1. Etymology

From the Greek “palaeo” for old and in reference to its morphology and similarity to asexual states linked to the fungal genus

Ophiocordyceps and from the Greek “kokkos” for scale and the Greek “phagos” for to eat, in reference to the fungus parasitizing a scale insect (Coccinea).

3.1.2. Type

Holotype specimen is deposited in the Poinar amber collection maintained at Oregon State University (B-Sy-18-holotype).

3.1.3. Locality and age

The fossil was collected from lignitic seams in sandstone–lime-stone deposits in the Hukawng Valley in Myanmar. The mine site was located on the slope of the Noiye Bum hill about a mile (1.5 km) SSW of the old Khanjamaw mine site and southwest of Maingkwang (26°20'N, 96°36'E). This site is newly mined (Cruickshank, personal communication) and has been designated as the “Noiye Bum 2001 Summit Site.” Paleontological findings from this site assigned its age to the Upper Albian of the Early Cretaceous (Cruickshank and Ko, 2003). On the basis of spectroscopic and anatomical evidence, the source of Burmese amber from this site was determined to be a member of the Araucariaceae (Poinar et al., 2007). The amber piece containing the parasitized fossil male scale insect is 8 mm long, 6 mm wide and 2 mm deep.

3.1.4. Commentary

The male scale insect is partly deteriorated as a result of the infection, fossilization and age. Its size, horizontal lateral rows of simple eyes, 10-segmented antennae and two pairs of long caudal setae align it with the family Albicoccidae Koteja, which occurs in Burmese amber (Koteja, 1999, 2000, 2004). The morphological characterization of the fossil fungus is consistent with the asexual genera *Hirsutella* and *Hymenostilbe*, asexual or mitotic spore-producing states linked to the meiotic genus *Ophiocordyceps* (Sung et al., 2007a). The genus *Hymenostilbe* includes 13 described species that are primarily pathogens of wasps, ants and scale insects (Hodge, 2003). It is characterized by the production of a palisade of cylindrical to clavate spore-producing cells (conidiogenous cells) that are typically produced on a stroma and possess multiple (polyblastic) apical denticles, which give rise to mitotic spores (conidia). Species of *Hirsutella* are pathogens of a more diverse assemblage of insect orders (Hodge, 2003). Many species of *Hirsutella* produce stromata that are morphologically similar to *Hymenostilbe*, but species of *Hirsutella* most typically produces conidiogenous cells that are more or less discontinuously distributed (i.e., not in a hymenium-like layer) and that are basally inflated with relatively long and slender tapered necks. They produce one to few conidia that accumulate in a droplet at the tip of the neck and lack the denticulate morphology of *Hymenostilbe*. A close relationship between *Hymenostilbe* and *Hirsutella* has long been supported based on interpretations of morphology and alpha taxonomy (Hodge, 2003). For example, the asexual state of *Ophiocordyceps clavulata*, a pathogen of scale insects, has been described as both a *Hirsutella* (Petch, 1933) and a *Hymenostilbe* (1950), highlighting the intermediate morphology of some taxa. Molecular phylogenetic studies on Hypocreales (Sung et al., 2007a,b) support a restricted phylogenetic distribution and a close relationship between *Hirsutella* s.s. and *Hymenostilbe*, and demonstrated that many species of *Ophiocordyceps* are predominantly linked to *Hymenostilbe* and *Hirsutella* asexual states (Fig. 2). Numerous questions remain, however, regarding subgeneric relationships within *Ophiocordyceps*, including relationships among species of *Hirsutella* and *Hymenostilbe*. For example, while *Hymenostilbe* (marked with an asterisk in Fig. 2) is supported as a member of the *Ophiocordyceps* clade, its phylogenetic placement as a sister-group to all remaining *Ophiocordyceps* or nested as a more terminal lineage within *Ophiocordyceps* was not resolved with strong statistical support and both topologies remain viable

phylogenetic hypotheses (Supplementary Fig. 1). Therefore, while we can confidently link the fossil to *Ophiocordyceps* we can not ascribe it to any subclade of the genus and we use *P. coccophagus* as a crown calibration point for *Ophiocordyceps* clade (Fig. 2).

3.2. Phylogenetic analyses

Final alignment contained 4936 unambiguously aligned positions, 4595 of which were variable. Maximum likelihood (ML) analyses resulted in a tree of -101742.21 log-likelihood. All five 10,000,000 generation Bayesian analyses converged on almost the same log-likelihood within 3,000,000 generations (Supplementary Fig. 2) and a 50% majority-rule consensus tree was generated from one of the five analyses by excluding the initial 3000 trees. The Bayesian analyses inferred an almost identical topology to the ML tree, which is shown in Supplementary Fig. 1 with ML bootstrap proportion (BP) and Bayesian posterior probability (PP).

All of the major families in Hypocreales were strongly supported (Supplementary Fig. 1) and the overall topology for the higher relationships is nearly congruent with the results from the previous phylogenetic analyses (Castlebury et al., 2004; Spatafora et al., 2007; Sung et al., 2007a). The *Stachybotrys* clade is resolved as a basal lineage of Hypocreales and all other lineages form a monophyletic group with strong statistical support (BP = 73%, PP = 1.00 in Supplementary Fig. 1), including the sister-group relationships of Clavicipitaceae + Ophiocordycipitaceae (BP = 95%, PP = 1.00 in Supplementary Fig. 1) and Hypocreaceae + Cordycipitaceae (BP = 72%, PP = 0.96 in Supplementary Fig. 1). The remaining higher relationships among major families were also resolved as in previous studies (Castlebury et al., 2004; Spatafora et al., 2007; Sung et al., 2007a).

3.3. Dating and evolution of host affiliation in the hypocrealean fungi

The Cretaceous diversification of angiosperms fundamentally changed terrestrial ecosystems and affected the terrestrial biodiversity of other organisms. One of the best-documented examples is the intimate association of numerous lineages of arthropods with the radiation of angiosperms (Barrett and Willis, 2001; Farrell, 1998; Gaunt and Miles, 2002; Labandeira, 1997; Percy et al., 2004; Ramirez et al., 2007). Although the diversity of fungi is comparable to that of arthropods and the radiation of some lineages of fungi is undoubtedly associated with the radiation of angiosperms, fungal examples of the Cretaceous diversification have not been confidently addressed by molecular phylogenetic analyses.

Dating analyses in this study support the supposition that the ancestor of the hypocrealean fungi was at least present in the Early Jurassic (193 Mya with CI of 158–232, node 1; Fig. 2 and Table 1). Our results also indicate that the major hypocrealean familial lineages originated in the Late Jurassic and diversified in the Cretaceous (Fig. 2). Fossil evidence shows that significant angiosperm diversification occurred between 115 and 90 Mya (Lidgard and Crane, 1988; Wing and Boucher, 1998), with a 125 million-year-old water lily (*Nymphaeaceae*) representing the earliest unequivocal record for an angiosperm crown group (Friis et al., 2001; Soltis and Soltis, 2004; Crepet, 2008). Because the diverse monocot and eudicot lineages that radiated during this period modified terrestrial ecosystems and initiated interactions with other organisms (Farrell, 1998; Labandeira and Sepkoski, 1993), a Cretaceous diversification of the hypocrealean fungi is consistent with the diversification of the angiosperms and their associated biota.

The ancestral nutritional state of the hypocrealean fungi was inferred to be plant-based, followed by a shift to animal and fungal hosts (Fig. 2). Species of the *Stachybotrys* clade, *Bionectriaceae* and *Nectriaceae* are primarily saprobes of plants, including both

angiosperms and gymnosperms, and plant products and many species of the *Nectriaceae* also are causal agents of plant diseases (e.g., *Fusarium*) (Rossman et al., 1999). Since the ancestral character state at node 5 is equivocally reconstructed as either animal, fungal or plant, the earliest fungal–animal symbiosis in the hypocrealean fungi is thought to initiate at either crown node 5 or 7, for which age estimates at least date to 173 Mya (CI: 146–206) or 158 Mya (CI: 137–187), respectively. In addition, the evolution of fungal–animal symbioses of the hypocrealean fungi is characterized by the origin and diversification of three families, Clavicipitaceae, Cordycipitaceae and Ophiocordycipitaceae, with most of their species serving as arthropod pathogens (Fig. 2). Our age estimates support a Cretaceous diversification of numerous lineages within these three families and their fungal–arthropod interactions.

3.3.1. Cordycipitaceae

The crown age of the Cordycipitaceae is inferred to be at least the Early Cretaceous (131 Mya with CI of 104–163, node 13; Table 1) and its ancestral ecology is reconstructed as animal (57%) or fungal (34%) (Fig. 2). The early diverging lineages of the Cordycipitaceae comprise parasites of mushrooms and rust fungi. This family is a sister-group to the Hypocreaceae, which primarily includes parasites or saprobes of other fungi such as mushrooms and polypores (Rossman et al., 1999). The ancestral hosts of the crown group of the Cordycipitaceae and Hypocreaceae (node 6 in Fig. 2) are equivocally inferred to be animal (45%), fungi (34%) or plant (20%) and the polarities of the interkingdom host shifts are not clearly defined.

Our results, however, do support a Cretaceous diversification of several of the arthropod pathogen lineages of the Cordycipitaceae. The age estimate for the origin of fungal–arthropod symbioses in the Cordycipitaceae is at least in the Early Cretaceous (116 Mya with CI of 91–146, node 16). The majority of insect pathogens in the Cordycipitaceae attack Lepidoptera while a few infect Coleoptera (Kobayasi, 1941, 1982; Sung et al., 2007a). This host pattern can be explained by the association of Lepidoptera with the radiation of angiosperms during the Cretaceous. The earliest fossil record of the Lepidoptera is from the Early Jurassic (ca. 190 Mya) (Whalley, 1985). While moths and butterflies comprise one of the most successful lineages of phytophagous insects, they are also one of the most recently evolved insect orders. The molecular age estimates (Gaunt and Miles, 2002), coupled with the numerous fossil records (Labandeira and Sepkoski, 1993; Rasnitsyn and Quicke, 2002), support the Cretaceous and Early Tertiary radiations of the major lineages of the Lepidoptera, which is likely to be associated with the diversification of fungal–arthropod symbioses.

3.3.2. Clavicipitaceae

The archetype of the Clavicipitaceae is resolved as an animal pathogen and the age estimate of its crown node is at least 117 Mya (CI: 95–144, node 14) in the Early Cretaceous (Fig. 2 and Table 1). The Cretaceous diversification and cladogenesis of the family resulted in several subclades, representing multiple interkingdom host shifts among three major kingdoms of eukaryotes (animal, plant and fungi). The family includes the ergot and grass endophytes (e.g., *Claviceps*, *Balansia* and *Epichloe*) that were hypothesized to be derived from an ancestral animal pathogen through interkingdom host shifts (Spatafora et al., 2007). In addition, the family also includes the plant-associated genus *Shimizuomyces*, which parasitizes the seeds of *Smilax sieboldii* (*Smilacaceae*) (Kobayasi, 1981). These reconstructions favor independent, parallel origins of the two fungal–plant associations within the Clavicipitaceae, although a single origin of plant–fungal association with a reversal to animal pathogen at node 17 could not be unequivocally rejected.

The grass symbionts in the family are specifically associated with members of the spikelet clade (e.g., genera *Arundo*, *Bambusa*, *Oryza* and *Zea*) in the grass family Poaceae (Clay and Schardl, 2002; Diehl, 1950; Pazoutova, 2001), which are characterized by two-glumed spikelets with three-staminate flowers (GPWG, 2001). The crown node of grass symbionts is estimated here to be at least 81 Mya (CI: 60–105, node 18), suggesting that fungal–grass symbioses originated in the Late Cretaceous (Fig. 2). These results are contradictory with some molecular clock analyses (Bremer, 2002) in which the grass family is estimated to have originated around the K–T boundary based on the calibration of a 55 million-year-old fossil record (Crepet and Feldman, 1991). More recent fossil evidence, however, suggests that monocots and grasses maybe older. Pollen similar to modern grasses (Crepet et al., 2004) and grass and palm remains from dinosaur coprolites (Prasad et al., 2005) were recently documented from Maastrichtian deposits (66–74 Mya), and a 100 million-year-old grass-like fossil that possesses spikelets (bambusoid-type) from Burmese amber (Poinar, 2004) indicate that the grass clade may be considerably older. Additionally, Gandolfo et al. (2002) recently discovered a series of ~90 million-year-old flower fossils that are members of the extant monocot family Triuridaceae, and molecular studies (Bremer, 2000; Davies et al., 2004; Schneider et al., 2004) have estimated the origin of monocots to the early Cretaceous with many lineages of extant monocots dating to 80–100 Mya. The data presented here adds to a growing body of the literature that is consistent with an earlier origin of the grass family than currently captured by the fossil record.

In addition to plant-associated species, the Clavicipitaceae includes pathogens of arthropods and nematodes (Fig. 2). The Cretaceous radiation of the subclade includes species of *Metacordyceps*, which is characterized by intrakingdom host shifts among the invertebrate groups, Coleoptera, Hemiptera, Lepidoptera and Nematoda (Fig. 2). The reconstruction of the ancestral ecology of *Metacordyceps* is complicated by the diversity of hosts, although arthropod pathogens are the dominant ecology of the subclade. In contrast, species of the *Hypocrella* subclade (e.g., *Aschersonia* and *Hypocrella*) are only known as pathogens of scale insects and white flies (Fig. 2). The age estimates for the *Hypocrella* clade (node 17) is at least 75 Mya with CI of 54–98, correspond to the Late Cretaceous (Fig. 2 and Table 1). Their ancestral ecology is inferred to be pathogens of scale insects, supporting a long evolutionary history for the interactions between scale insects and fungi of the *Hypocrella* clade.

3.3.3. Ophiocordycipitaceae

The ancestral ecology of Ophiocordycipitaceae is unequivocally reconstructed as an animal pathogen (Fig. 2) and the age estimates of its crown group are at least in the Early Cretaceous (122 Mya with CI of 109–138, node 15). The largest genus within the family is *Ophiocordyceps*, which includes pathogens of arthropods. Although the monophyly of the genus receives less than 70% bootstrap proportion in these analyses (Supplementary Fig. 1), previous studies with sequence data from additional genes demonstrated that *Ophiocordyceps* is a monophyletic group with good statistical support (Spatafora et al., 2007; Sung et al., 2007b). Species of the genus parasitize insects in the orders Coleoptera, Hemiptera, Hymenoptera and Lepidoptera (Fig. 2) and collectively represent the most diverse clade of arthropod pathogens in Hypocreales (Sung et al., 2007a). Most species parasitize larval, nymphal and pupal stages of subterranean and wood-inhabiting insects, but notable exceptions attacking adult hosts in exposed settings exist (e.g., *O. unilateralis* on adult ants). This pattern contrasts with species in the Cordycipitaceae that primarily parasitize arthropods in leaf litter. Because the Cretaceous radiation of pathogens of arthropods appear to be intimately related to the diversification of angio-

sperms (Fig. 2), the nearly simultaneous origins we observe for these two families indicate that their diversification may have occurred in different niches, i.e., litter-inhabiting hosts for the Cordycipitaceae versus soil and wood-inhabiting hosts for the Ophiocordycipitaceae (Sung et al., 2007a).

Although the dominant hosts of the Ophiocordycipitaceae are arthropods, other patterns of host affiliation represented in this family include parasitism of nematodes and fungi (Fig. 2). Nematode parasitism in the family is primarily represented by asexual forms closely related or linked to *Cordyceps sensu lato* and other arthropod pathogenic taxa (e.g., *Haptocillium* and *Paecilomyces*) (Zare and Gams, 2001). The sexual forms of nematode parasites often parasitize arthropods (e.g., genus *Podocrella*) and there is increasing evidence for the presence of disparate host affiliations between sexual and asexual forms of the same species (Chaverri et al., 2005). This trend is also found in distantly related species in the Clavicipitaceae (Fig. 2) where the asexual form *P. chlamydosporia*, which is a nematophagous species, is linked to *Metacordyceps chlamydosporia*, a parasite of mollusc eggs (Sung et al., 2007a; Zare et al., 2001). Interestingly, nematophagous species in Ophiocordycipitaceae appear to differ from nematophagous species of the Clavicipitaceae in host preference (active free-living nematodes versus sedentary nematode cysts and eggs, respectively) and some have proposed that nematode parasitism may have evolved numerous independent times from fungal–arthropod symbioses (Zare and Gams, 2001; Zare et al., 2001), a finding consistent with our results (Fig. 2).

Species of Ophiocordycipitaceae that parasitize members of the fungal genus *Elaphomyces* (Kobayasi and Shimizu, 1960; Mains, 1957) are classified in the genus *Elaphocordyceps*, which also includes parasites of soil-inhabiting cicada nymphs (e.g., *E. inegoensis*) and wood-inhabiting beetle larvae (e.g., *E. subsessilis*) (Sung et al., 2007a). Our results support an interkingdom host shift from animals to fungi in the Late Cretaceous (70 Mya with CI of 56–84, node 20; Fig. 2). This estimate is older than the previous estimate (43 Mya) based on nucleotide substitutional rates in small subunit rDNA (Nikoh and Fukatsu, 2000). While arthropod hosts of the genus include two distantly related taxa (Coleoptera and Homoptera), all fungal parasites are restricted to a single host genus, i.e., *Elaphomyces*, although numerous other truffle taxa co-exist in these environments. Ancestral reconstructions of internal nodes of *Elaphocordyceps* favor fungal ancestral hosts with multiple parallel reversals to animal hosts (Fig. 2). The ecological, morphological and molecular evidence is consistent, therefore, in hypothesizing subsequent reversals from truffles to beetle and cicada hosts (Sung et al., 2007a) (Fig. 2).

4. Concluding remarks

We describe the oldest fossil evidence of a known fungal parasite of an animal, *P. coccophagus*, and use it to calibrate the multi-gene phylogeny of Hypocreales. By doing so we provide evidence for parallel Cretaceous diversification of insect pathogens of three families of Hypocreales, namely Clavicipitaceae, Cordycipitaceae and Ophiocordycipitaceae. We provide additional evidence that pathogenicity of animals–arthropods in particular—is an ancestral character for Clavicipitaceae and Ophiocordycipitaceae, and possibly for Cordycipitaceae, and that the current phylogenetic distribution of host affiliation is the product of frequent intra- and interkingdom host shifts (Fig. 2). Based on an estimate from monographic studies (Kobayasi, 1941, 1982; Mains, 1958), more than 75% of the hypocrealean arthropod pathogenic fungi are associated with insects in the orders Coleoptera, Hemiptera and Lepidoptera, which collectively represent the greatest diversity of phytophagous insects (Rasnitsyn and Quicke, 2002). The radiations of the phytophagous insects are intimately associated with the angio-

sperm diversification in the Cretaceous (Farrell, 1998; Gaunt and Miles, 2002) and these results support parallel Cretaceous diversifications of many lineages of insect pathogens in three predominantly arthropod pathogenic families of Hypocreales. Today, more than 25% of all extant arthropod species are phytophagous insects and occupy 50% of the biomass and 75% of the individuals in temperate deciduous forests (Rasnitsyn and Quicke, 2002). Therefore, the diversity of arthropod pathogens of Hypocreales may be directly related to ecological dominance of the phytophagous insects and their coevolutionary interactions with angiosperms during the Cretaceous.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.ympev.2008.08.028.

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